CLASSIFICATION

NON BINARY CLASSIFICATION

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NON BINARY CLASSIFICATION

The following concepts will be introduced:

- ✓ **NON BINARY CLASSIFICATION**
  - MULTI-CLASS
  - MULTI-LABEL
- ✓ **ONE-VS-ALL CLASSIFICATION**

Non Binary Classification

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It may be the case that you are asked to solve a **NON BINARY CLASSIFICATION** problem, i.e., a classification problem where **THE CLASS ATTRIBUTE** is:

Types of Eczema

Atopic
Contact
Dyshidrotic
Nummular
Neurodermatitis
Seborrheic

Non Binary Classification

Multi-class Classification

a single class value is allowed

no one, one or more class values are allowed

Multi-label Classification

Taste

Awful
Poor
Ok
Good
Great

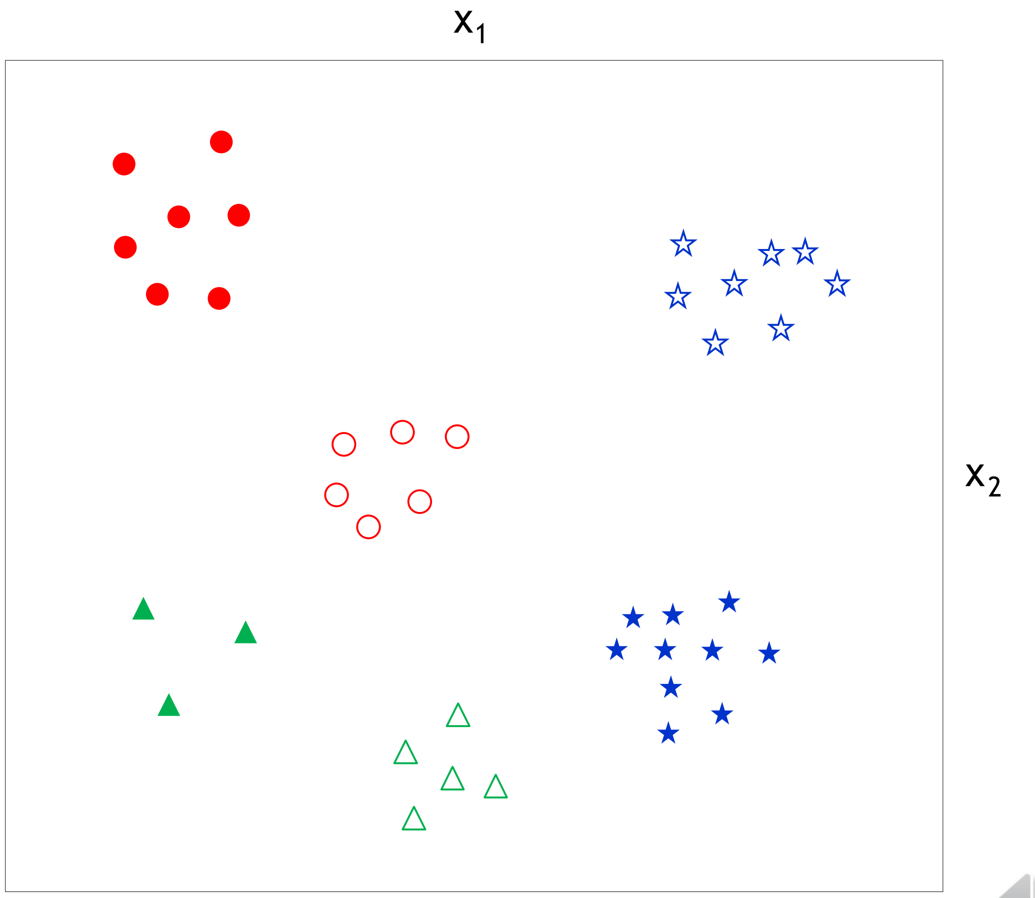
Ranking Problem

A **Non Binary Classification** problem is typically solved by applying the **One-Vs-All** (One-Against-All) transformation:

Set of Binary Classification Problems

Types of Eczema

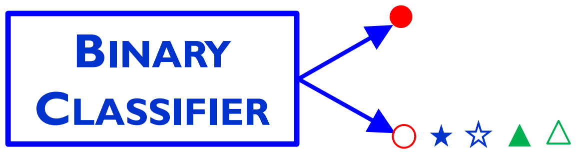
Types of Eczema		Set of Binary Classification Problems	
Atopic	→	Atopic or not?	●
Contact	→	Contact or not?	○
Dyshidrotic	→	Dyshidrotic or not?	★
Nummular	→	Nummular or not?	☆
Neurodermatitis	→	Neurodermatitis or not?	▲
Seborrheic	→	Seborrheic or not?	△

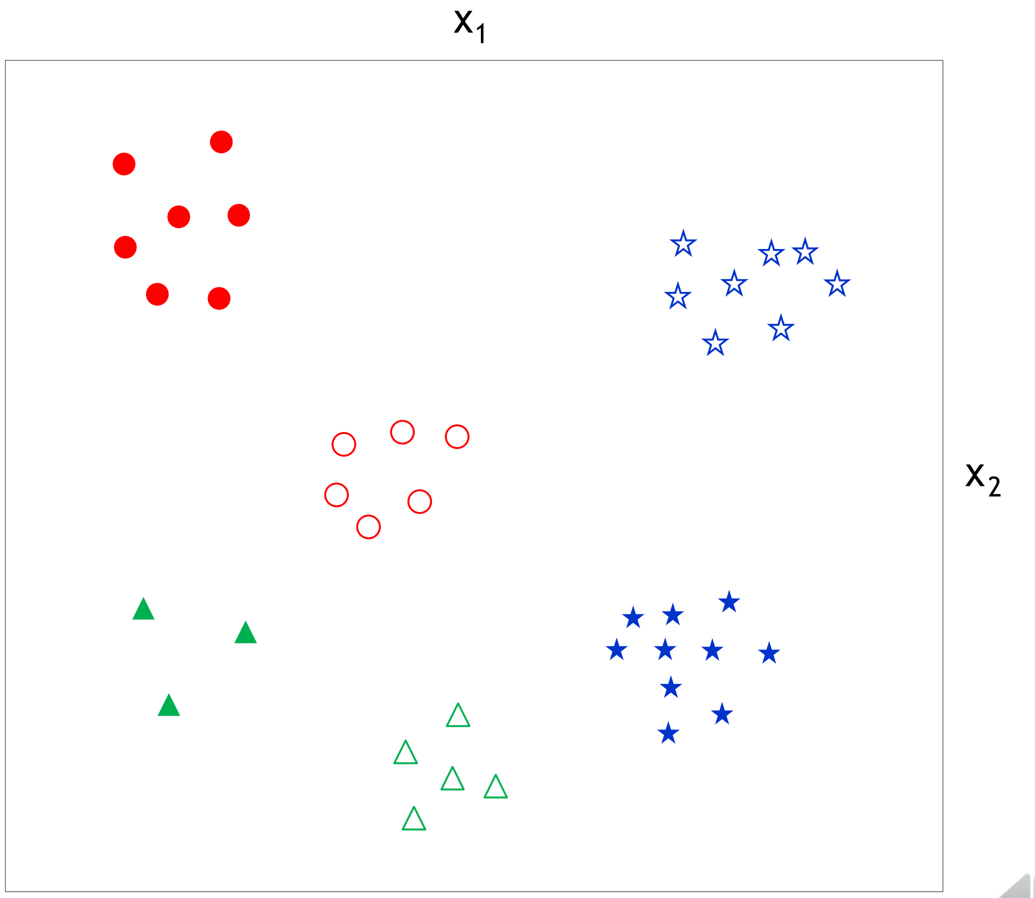


A **Non Binary Classification** problem is typically solved by applying the **One-Vs-All** (One-Against-All) transformation:

One-Vs-All

Develop a **Binary Classifier** for each value of the class.



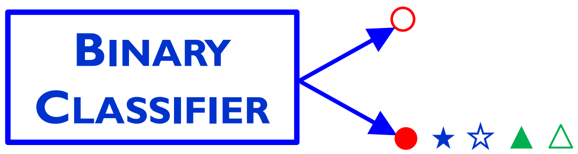


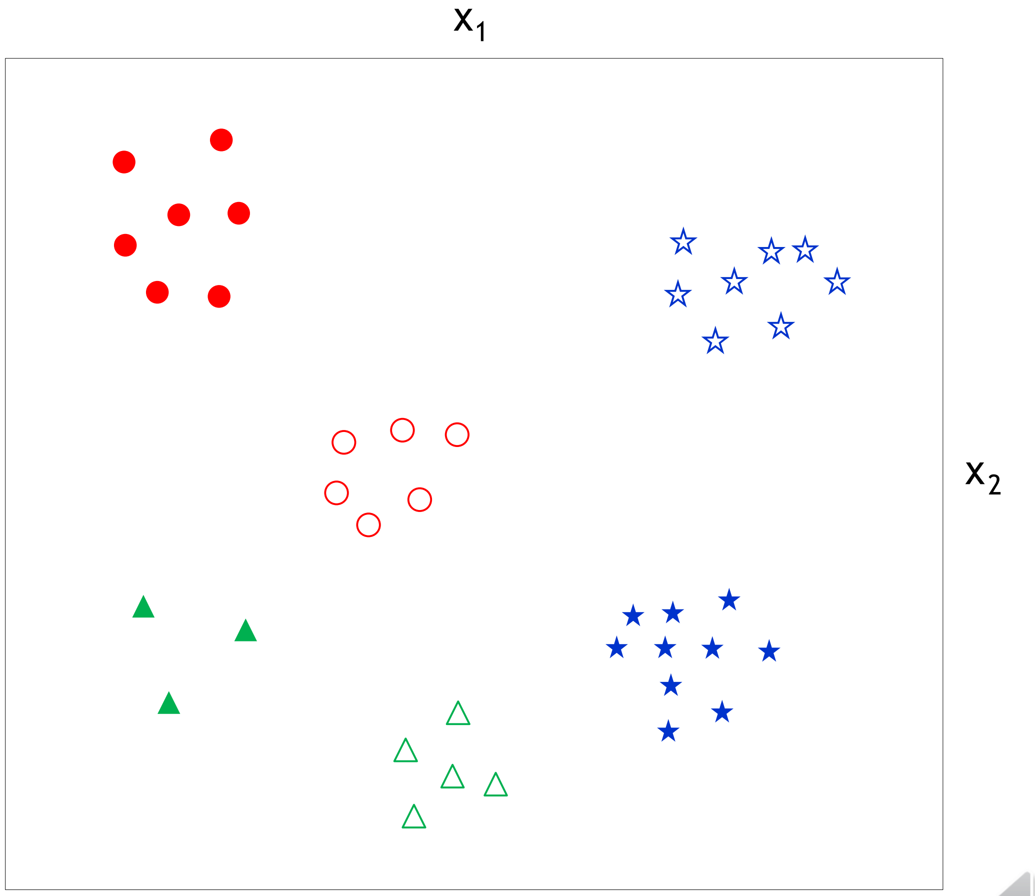
NON BINARY CLASSIFICATION: MULTI-CLASS 2

A **NON BINARY CLASSIFICATION** problem is typically solved by applying the **ONE-VS-ALL** (One-Against-All) transformation:

ONE-VS-ALL

Develop a **BINARY CLASSIFIER** for each value of the class.



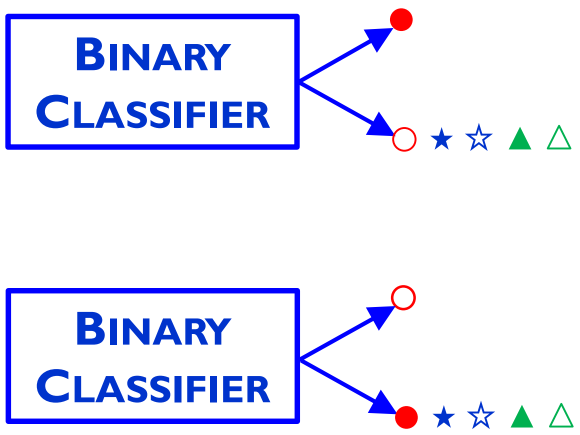


NON BINARY CLASSIFICATION: MULTI-CLASS 3

A **NON BINARY CLASSIFICATION** problem is typically solved by applying the **ONE-VS-ALL** (One-Against-All) transformation:

ONE-VS-ALL

Develop a **BINARY CLASSIFIER** for each value of the class.



x_1, x_2 ●

MULTI-CLASS CLASSIFICATION

●	**0.78**
○	0.12
★	0.05
☆	0.01
▲	0.03
△	0.01

6 VALUES FOR THE CLASS, 6 BINARY CLASSIFIERS

SET OF BINARY CLASSIFIERS

NON BINARY CLASSIFICATION: MULTI-CLASS 4

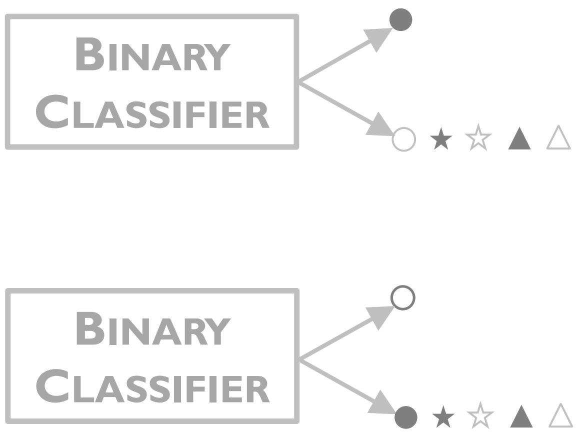
A **NON BINARY CLASSIFICATION** problem is typically solved by applying the **ONE-VS-ALL** (One-Against-All) transformation:

ONE-VS-ALL

Develop a **BINARY CLASSIFIER** for each value of the class.

THRESHOLD

0.5



x_1, x_2 ● ☆

MULTI-LABEL CLASSIFICATION

●	0.51
○	0.12
★	0.15
☆	0.65
▲	0.23
△	0.08

6 VALUES FOR THE CLASS, 6 BINARY CLASSIFIERS

SET OF BINARY CLASSIFIERS